

Kubernetes and Cloud Native Associate - KCNA

Cloud Native Architecture

Cluster Autoscaler

In this article, we explore the Cluster Autoscaler and its crucial role in optimizing resource allocation within a Kubernetes cluster. Previously, we discussed how pods consume resources and how the Kubernetes scheduler assigns these pods to nodes with enough available capacity. However, as more pods are deployed, available resources become scarce. Once nodes run out of sufficient resources, pods enter a pending state, and the Cluster Autoscaler steps in to automatically scale the cluster by adding new compute instances.

The Cluster Autoscaler integrates seamlessly with the underlying infrastructure provided by your cloud provider. Each supported provider has its own specific requirements and capabilities. For an up-to-date list of compatible cloud providers, please refer to the [Kubernetes Cluster Autoscaler documentation](#). For example, the diagram below displays the supported cloud providers:

FAQ/Documentation

An FAQ is available [HERE](#).

You should also take a look at the notes and "gotchas" for your specific cloud provider:

- AliCloud
- AWS
- Azure
- BaiduCloud
- BizflyCloud
- Brightbox
- CherryServers
- Civo
- CloudStack
- ClusterAPI
- DigitalOcean
- Exoscale
- Equinix Metal
- External gRPC
- Hetzner
- HuaweiCloud
- IonosCloud
- Kamatera
- Linode
- Magnum
- OracleCloud
- OVHcloud
- Rancher
- Scaleway
- TencentCloud
- Vultr

<https://github.com/kubernetes/autoscaler/tree/master/cluster-autoscaler>



Note

On [Google Cloud](#), you can easily configure the Cluster Autoscaler by setting autoscaling options during the creation of your Kubernetes cluster.

To illustrate, when creating a cluster on Google Cloud, you can specify a minimum of 3 nodes and a maximum of 10 nodes. This configuration ensures that when pods remain in a pending state due to insufficient resources, the Cluster Autoscaler will automatically increase the number of nodes to meet the demand. Conversely, it will scale down the cluster when fewer resources are needed.

The following command demonstrates how to create a cluster with the Cluster Autoscaler enabled on Google Cloud:

```
gcloud container clusters create my-cluster --cluster-autoscaler=min-nodes=3,max-
```



Warning

Configuration settings can vary significantly between cloud providers. Always consult your specific cloud provider's documentation for precise instructions.

Below is a quick summary of key aspects of the Cluster Autoscaler:

Aspect	Description	Example Command/Reference
Resource Scaling	Automatically adds nodes when pods are pending due to resource shortages	<code>`gcloud container clusters create - -cluster-autoscaler`</code>



Beta



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For further reading, check out:

- [Kubernetes Documentation](#)
- [Cloud Provider Documentation](#)



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